

Tier	Function / Component	Manufacturer	Manufacturer Part Number	Description / Specification	Quantity	Unit Cost	Total Cost	Interface
2	Linux application / graphics SoC	AMD	XQZU3EG-15FRA484I	XQ Zynq UltraScale+ MPSoC; application processors, GPU, FPGA fabric	1			484-ball FCBGA; DDR4; QSPI; eMMC; DisplayPort; Ethernet
2	RTOS / supervisory MCU	Texas Instruments	SSL520216ASPGEMEP	TMS570LS20216-EP Hercules MCU; lockstep Cortex-R4F	1	55.49	55.49	144-LQFP; SPI; UART; JTAG; GPIO
2	ARINC 429 transceiver	Holt Integrated Circuits	HI-35930PQTF	Dual ARINC 429 receiver and single transmitter IC	2			SPI to MCU; ARINC 429 differential I/O
2	DDR4 system memory	ISSI	IS46QR16512A-075VBLA2	8-Gbit x16 DDR4 SDRAM	2	55.15	110.3	96-TFBGA; DDR4
2	QSPI boot flash	Micron	MT25QL512AB8E12-0AUT	512-Mbit Quad-SPI NOR flash	1	61.89	61.89	24-TBGA; QSPI; 3.3 V
2	Main nonvolatile storage	Swissbit	SFEM020GB2ED1TB-A-CE-11P-STD	20 GB eMMC storage	1	164.85	164.85	153-BGA; eMMC
2	SoC power-management IC	Texas Instruments	TPS65086401RSKT	Multi-rail PMIC for processor / FPGA power rails	1	10.37	10.37	64-VQFN; I2C; power rails
2	Reset supervisor	Texas Instruments	TPS3808G01MDBVTEP	Adjustable voltage supervisor / reset controller	1	6.31	6.31	SOT-23-6; reset
2	Independent watchdog	Texas Instruments	TPS3431SDRBR	Window watchdog timer	1	2.08	2.08	8-SON; digital watchdog
2	Gigabit Ethernet PHY	Texas Instruments	DP83867IRPAPR	10/100/1000 Ethernet physical-layer transceiver	1	7.23	7.23	RGMIIL to SoC; Ethernet differential pairs
2	microSD ESD protection	Texas Instruments	TPD6E05U06RVZR	6-channel low-capacitance ESD protection array	1	1.71	1.71	SD signal lines
2	microSD socket	Molex	104031-0811	Surface-mount microSD memory-card connector	1	2.36	2.36	SD/MMC
2	MFD LCD panel	SAEF	SFTO1330SM-7208A3CT2	13.3-inch 1920x1080 high-brightness display	1	95	95	eDP
3	SoC reference clocks	TBD	TBD	Reference oscillator / clock network required by the MPSoC	1			Clock inputs
3	JTAG / programming connector	TBD	TBD	Programming and hardware-debug connector	1			JTAG
3	Display connector / backlight support	TBD	TBD	Connector and support circuitry for final LCD panel	1			eDP / backlight
3	Aircraft external I/O connector	TBD	TBD	Rugged external connector for aircraft data and power	1			Aircraft I/O
3	Decoupling / termination passives	TBD	TBD	Required decoupling, pull-up, pull-down, bias, and termination components	1			PCB passives